

National College of Ireland

Project Submission Sheet

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Student ID:
Programme: Msc in Cloud Computing **Year:** 2026-2027
 Research in Computing (MSCCLOUD1JAN26I)
Module:
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Lecturer:
Submission Due Date: Friday 12th of June 2026
Project Title: Lightweight Fault detection and Localaization in AWS Serverless
 Microservices using CloudWatch and X-Ray

 1440
Word Count:

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Signature: Yashaswini Penumarthi
12-06-2026
Date:

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AI Acknowledgement Supplement

Research in Computing

Lightweight Fault Detection and Localisation in AWS Serverless Microservices Using CloudWatch and X-Ray

Your Name/Student Number	Course	Date
Yashaswini Penumarthi	Research in Computing	12-06-2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

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Tool Name	Brief Description	Link to tool

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[Insert Tool Name]	
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Additional Evidence:

[Place evidence here]

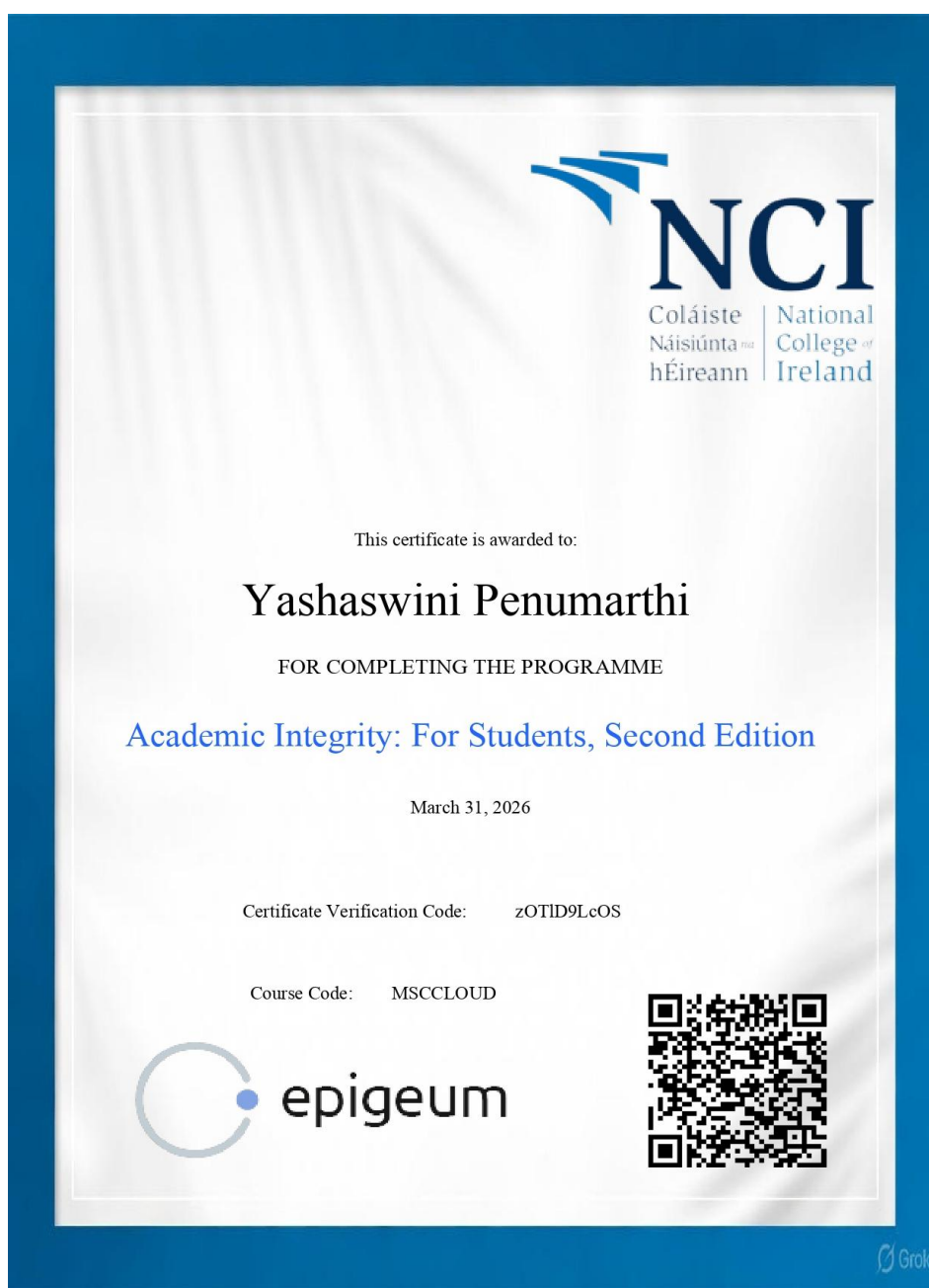
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Lightweight Fault Detection and Localisation in AWS Serverless Microservices Using CloudWatch and X-Ray

Yashaswini Penumarthi

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MSc in Cloud Computing

12th June 2026

1 Research Problem Background

In fact, timely fault detection and accurate fault localization are key concerns in observability, and in cloud-native systems, minor failures can cascade and affect the entire system as each service depends on others; Xing et al. (2025) propose a dual-channel deep-learning framework for detecting anomalies and localizing the root-cause service within the microservice telemetry, achieving strong detection accuracy ($F_1 \approx 93\%$) and localization accuracy (Top-3 $\approx 93\%$). But achieving that precision requires hours of training, a few gigabytes of memory and a significant telemetry set of labeled data, which is something that the smallest and newest serverless services are least equipped to provide.

The gap in practice is what is most often desired: low overhead observability (small, newly deployed serverless services operating at tight cost limits) is precisely what is not labelled history and compute by these high accuracy detectors. This same tension is echoed in the work of Zhang and Yang (2026) who argue that using a multitude of heterogeneous monitoring signals brings with it feature redundancy and modelling complexity, and in the work of Zhang et al. (2026) who report that while graph- and LLM-based localisation is promising in smaller systems, it suffers from scalability constraints in larger systems. Combined they validate an unresolved challenge: If you want high accuracy microservice fault detection and localisation, it's too heavy and too data-dependent for small, cost-constrained, newly-deployed, serverless workloads.

2 Research Question

How accurately, and at what monitoring overhead, can a lightweight CloudWatch- and X-Ray-based observability method detect and localise injected faults in AWS serverless microservices, compared with heavy dual-channel deep-learning detection-and-localisation models?

3 Justification

The importance of observability is that undetected, or slowly localised faults will turn into downtime and cost quickly, and the most accurate detectors published rely on a deep model that must be trained for hours with a significant amount of labelled history, memory and powerful machines that small and newly deployed serverless services do not have. This study explores how much non-zero- or even high-accuracy dual-channel fault detection and localisation can be achieved using such telemetry, with little or no historical training, and at what cost in terms of monitoring, pushing it towards the lightest configuration possible that is suitable for an individual small serverless service to run. The question can be answered in a modest, self-contained manner: a single order-processing microservice, with API Gateway, Lambda and DynamoDB CloudWatch metrics and logs created and emitted, and the detection logic, which is based on threshold and deviation rules over the metrics and the trace-dependency ranking from X-Ray, can be run on one machine, and the faults are injected with a controlled script, not in production. The evidence comprises detection precision, recall, F_1 and detection delay, reported alongside the monitoring overhead and estimated cost; as only synthetic workloads and data generated by the synthetic workload is involved, no personal data is processed, and resource use can be openly reported.

4 Specific Items to Be Addressed

1. *Observability signals*: examine the CloudWatch metrics and logs and X-Ray traces a serverless microservice emits and their usefulness for detecting and localising faults.
2. *Detection approach*: Develop a lightweight technique to mark anomalies using threshold rules and deviation rules on metrics, and rank the most probable underlying service using X-Ray trace dependencies, requiring very little or no training in the past.
3. *Fault scenarios*: study representative faults (timeouts, dependency failures, throttling and elevated latency) introduced under controlled conditions and increasing load.
4. *Detection and localisation accuracy*: measure detection precision, recall, F_1 and detection delay, and root-cause localisation Top- k accuracy and mean rank.
5. *Monitoring overhead*: quantify the log and trace volume, added latency, compute requirement and estimated cost the approach incurs.
6. *Trade-off and guidance*: Investigate the accuracy-overhead relationship with respect to high-performance dual-channel deep learning detection and localization models and provide a basis for small serverless services with respect to observability.

Bibliography

- Xing, S., Wang, Y., and Liu, W. (2025). Multi-dimensional anomaly detection and fault localization in microservice architectures: A dual-channel deep learning approach with causal inference for intelligent sensing. *Sensors*, 25(11):3396. doi: 10.3390/s25113396.
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- Zhang, W., Yang, Z., Peng, F., Zhang, L., Chen, Y., and Chen, R. (2026). GALR: Graph-based root cause localization and LLM-assisted recovery for microservice systems. *Electronics*, 15(1):243. doi: 10.3390/electronics15010243.